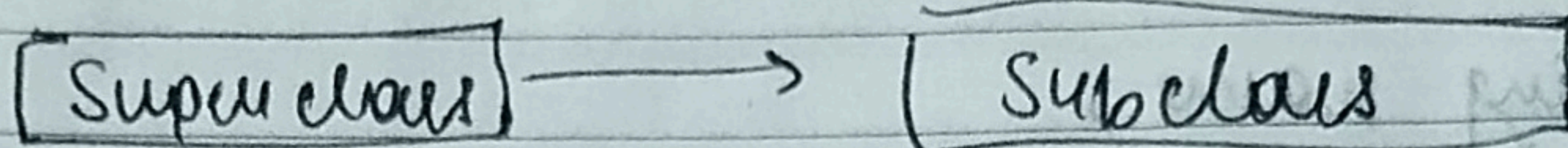
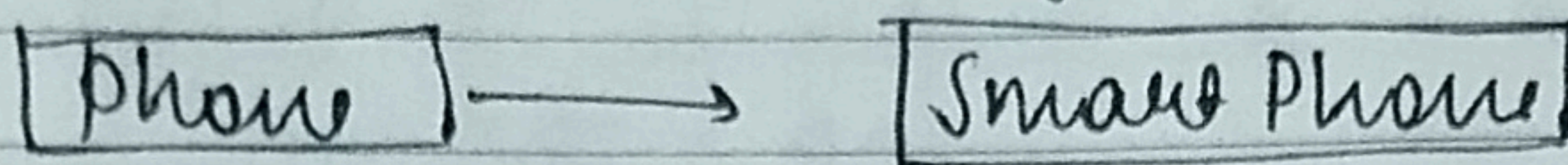


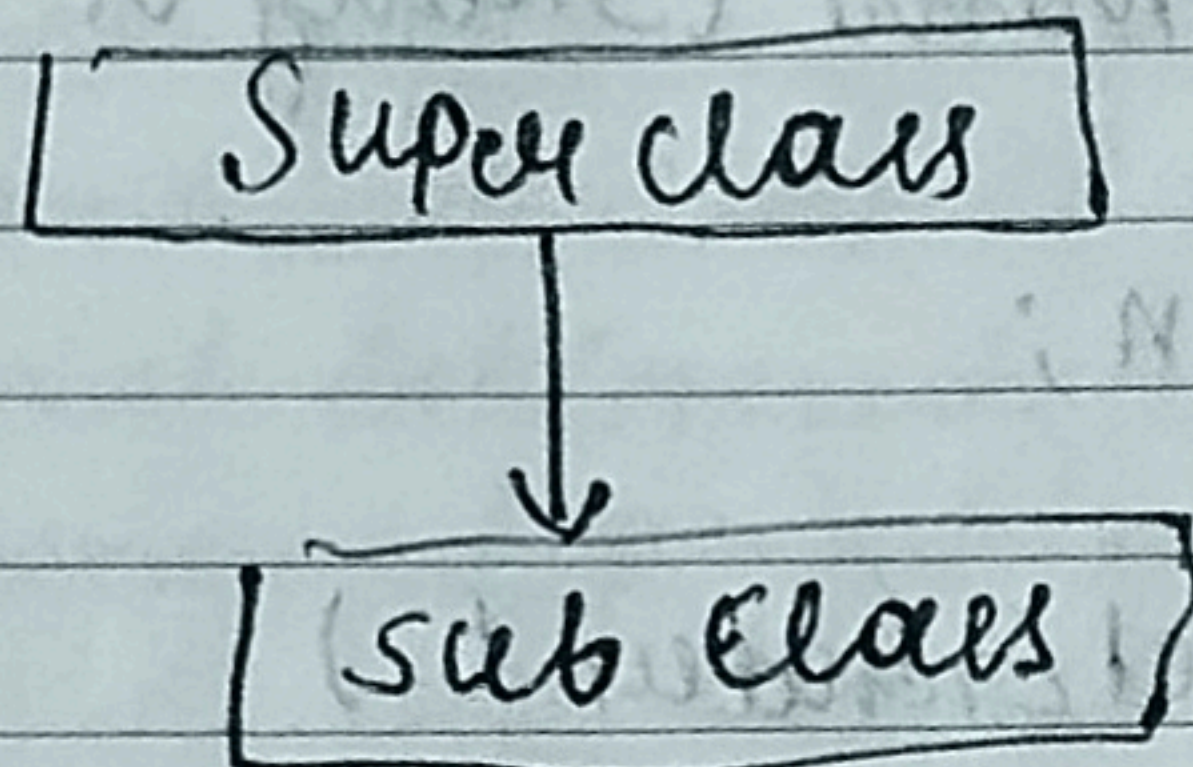
Chapter → 10 Inheritance

Inheritance is used to borrow properties & method from an existing class.



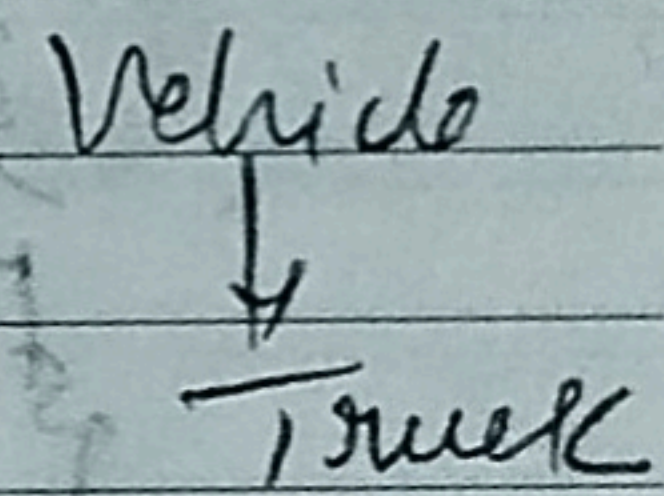
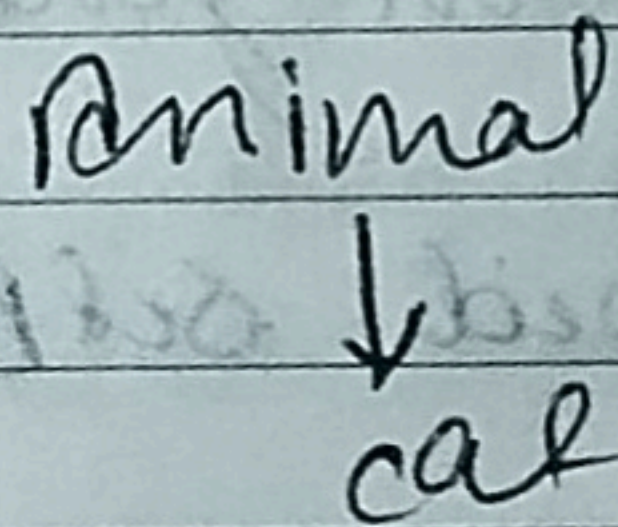
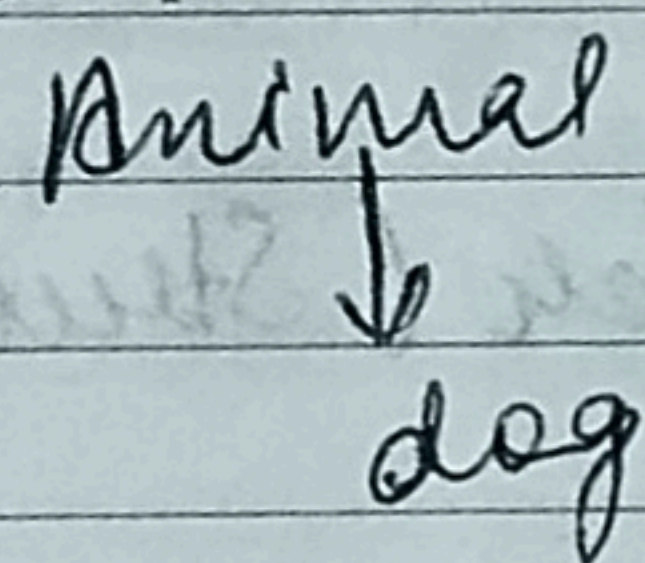
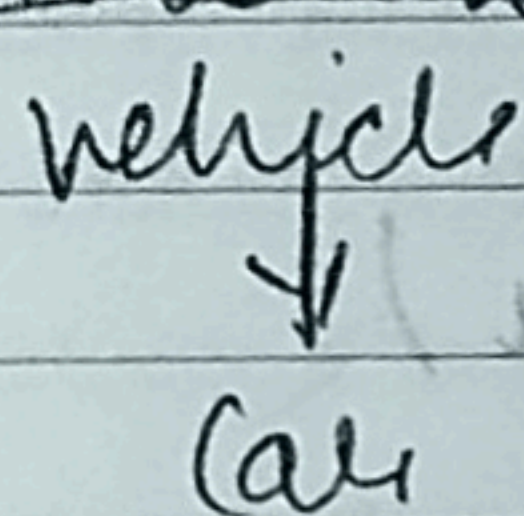
Declaring Inheritance in Java

Inheritance in Java is declared using extends keyword



⇒ Subclass extends the superclass

More Examples:



Quick Quiz:

Create a class Animal and Derive another class Dog from it.

```
class Animal
```

```
{
```

```
    String name;
```

```
    String gender;
```

```
    public String getName ()
```

```
    {
```

```
        return name;
```

```
    }
```

```
    public Stringvoid setName (String n)
```

```
    {
```

```
        name = n;
```

```
    }
```

```
    public String getGender ()
```

```
    {
```

```
        return gender;
```

```
    }
```

```
    public void setGender (String gen)
```

```
    {
```

```
        gender = gen;
```

```
    }
```

```
}
```

```
class Dog extends Animal
```

```
{
```

```
    int age;
```



```
public int getAge ()
```

```
{
```

```
    return age;
```

```
}
```

```
public void setAge (int age)
```

```
{
```

```
    this.age = age;
```

```
}
```

```
}  
public class INHERITANCE
```

```
{
```

```
    public static void main (String [] args)
```

```
{
```

```
        ANIMAL animal = new ANIMAL ();
```

```
        animal.setName ("Husky");
```

```
        System.out.println (animal.getName());
```

```
        animal.setGender ("Male");
```

```
        System.out.println ("animal.getGender());
```

```
        Dog dog = new Dog();
```

```
        dog.setAge (11);
```

```
        System.out.println (dog.getAge());
```

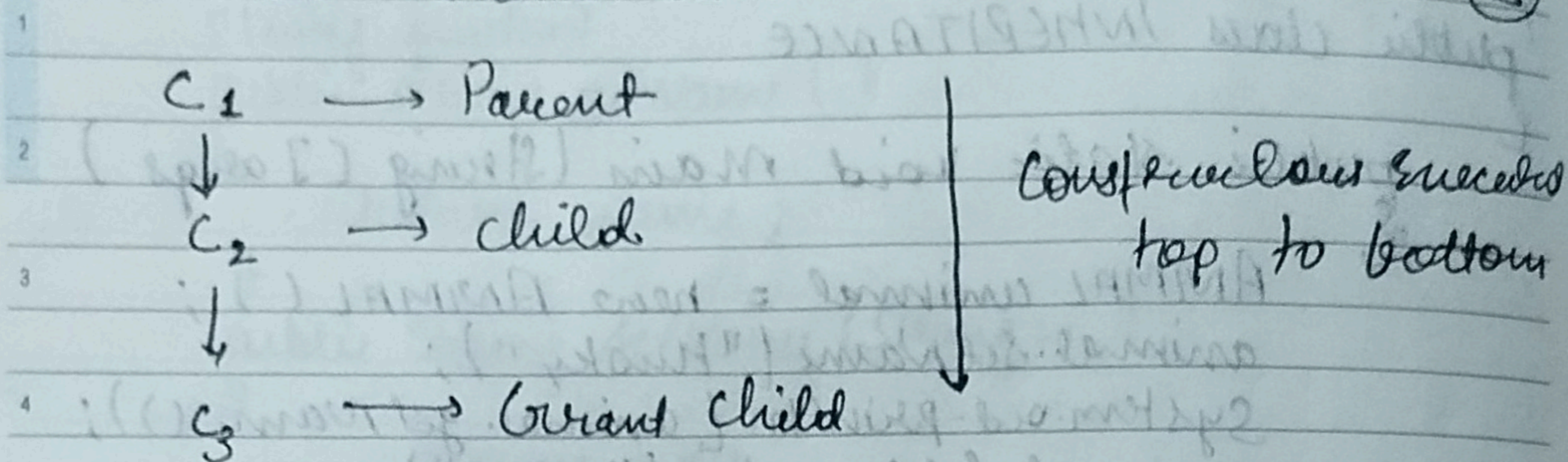
```
}
```

```
}
```

2

Constructors in Inheritance

- When a derived class is extended from the base class, the constructor of the Base class is executed first, followed by the constructor of the derived class.
- For the following Inheritance hierarchy, the constructors are executed in the order ① → ② → ③



Constructors during Constructor Overloading

When there are multiple constructors in the parent class, the constructor without any parameter is called from the child class.

If we want to call the constructor with parameter from the parent class, we can use super keyword.

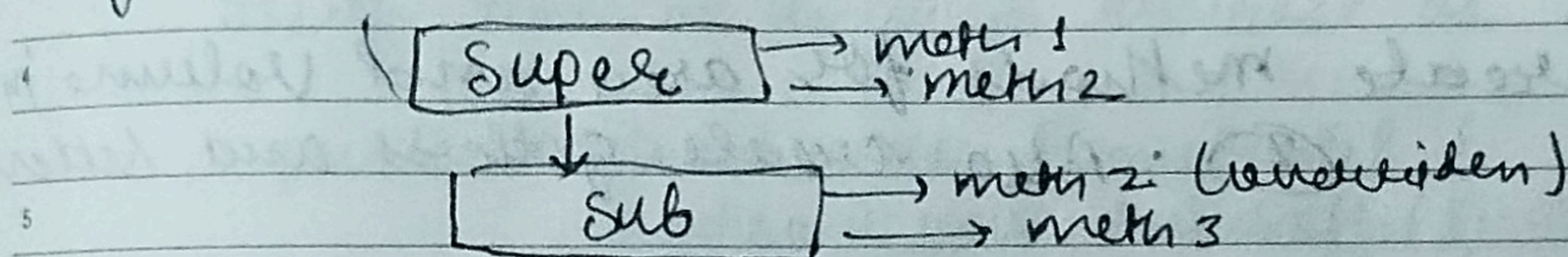
`Super(a,b);`

Method Overriding

if the child class implements the same method present in the parent class again, it is known as method overriding.

when an object of subclass is created and the overridden method is called, the method which has been implemented in the subclass is called & it is executed.

Dynamic Method Dispatch



Scenario 1 → Super obj = new Sub () → allowed
 obj. meth2 () → allowed
 obj. meth3 () → not allowed

Scenario 2 → Sub obj = new Super () → not allowed

SUN 02